Gona Sanjana

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.55
Sri Chaitanya Junior Kalasala	Class XII (Telangana Board of Intermediate Education)	2023	98.40%
Sri Chaitanya High School, Gaddiannaram, Hyderabad	Class X (The Board of Secondary Education, Telangana)	2021	10.00%

EXPERIENCE

VISA Inc
Software Engineering Intern

- **Built** an end-to-end AI-powered Intelligent File Parser using LangGraph agents to map heterogeneous partner files into a standardized 11-field schema, replacing rigid format-specific fraud-data parsers.
- **Designed** a multi-agent parsing pipeline with Parser, Validator, Repair, and Supervisor agents, combining LLM-based extraction with deterministic validation, bounded retries, confidence scoring, and prompt-injection safeguards.
- **Engineered** the production data pipeline with Visa Kafka over secure mTLS, idempotent consumers, ordered event processing, SQLite persistence, and Docker/Podman deployment on AMD64 VMs.
- **Built** a React-based upload/results UI, live APM dashboard, VSFU malware scanning, automated frontend/backend tests, and email-based parsing-report workflow.

Research Assistant
Prof. Sathya Peri, IIT Hyderabad

- **Investigated** lock-free and wait-free synchronization by designing concurrent shared-array algorithms in C++20 and benchmarking throughput and p99 latency under high contention.

PROJECTS

Centralized Comms Library - AI-Powered Content Platform | *Source Code*

- **Established** a full-stack TypeScript monorepo using React, Express, and Prisma to centralize content authoring, review, and publishing across teams.
- **Streamlined** content operations with a lifecycle engine, granular RBAC, co-author workflows, and quorum-based approvals.
- **Integrated** Elasticsearch with vector embeddings for semantic search and an AI tutor, containerizing the platform with Docker Compose across PostgreSQL, MongoDB, Node API, and nginx.

VMDT - Verifiable Multipath Decentralized Transport Protocol | *Source Code*

- **Designed** a novel transport protocol combining Shamir's Secret Sharing (SSMS), AES-256-CTR encryption, and multipath routing for secure peer-to-peer communication.
- **Built** a reliable UDP data-transfer layer with custom packet formats, sequence numbers, acknowledgments, and retransmission mechanisms for fault-tolerant delivery.
- **Engineered** a decentralized cluster-based architecture with dynamic share-pool management supporting k-out-of-n reconstruction for high availability.

Multi-Write - Lock-Free & Wait-Free Concurrent Shared Array | *Research · Prof. Sathya Peri*

- **Investigated** lock-based vs. non-blocking synchronization by building four thread-safe shared-array designs in C++20: global lock, fine-grained lock, lock-free snapshot CAS, and wait-free.
- **Achieved** wait-free progress via per-thread announcement slots and a propose-and-help protocol that bounds retries, reducing p99 multi-write tail latency under high contention.
- **Conducted** benchmark sweeps across read-heavy, mixed, and multi-write-only workloads, measuring throughput, p99 latency, and peak memory with correctness verified through concurrent stress tests and ThreadSanitizer.

SKILLS

- **Programming Languages:** C, C++, RISC-V, JavaScript, Bash
- **Web Technologies:** HTML, CSS, React, Next.js, Astro.js, Tailwind CSS, Bootstrap, Node.js, Express.js
- **Databases:** MySQL, PostgreSQL
- **Other Tools:** Git, GitHub, LaTeX
- **DevOps:** Docker, Vercel, Linux, AWS, Bash Scripting

POSITIONS OF RESPONSIBILITY

Inter IIT Cultural Meet 7.0 & Dancing Club, IIT Hyderabad | *Representative & Core Member*

- **Represented** IIT Hyderabad at Inter IIT Cultural Meet 7.0, choreographing and performing in 15+ cultural events, leading creative direction for 3 major performances, and training 25+ members in contemporary dance techniques.

Media Council, IIT Hyderabad | *Co-ordinator, Web Team*

- **Coordinated** development of the student web platform, managing timely content updates for 20+ events and providing technical support to 500+ students, resulting in 40% improved user engagement.

RELEVANT COURSES

- **Mathematics & Statistics:** Calculus, Discrete Mathematics, Probability and Random Variables, Introduction to Statistics, Linear Algebra, Transformation Techniques
- **Theory & Algorithms:** Data Structures, Algorithms, Theory of Computation, Compilers
- **Systems & Networks:** Operating Systems, Computer Architecture, Database Management Systems, Networks, Software Development Fundamentals
- **Specialized:** Introduction to Programming, Machine Learning, Software Engineering